

Lab Program 5(a): Installing Python, PyTorch, and Uploading Files to Google Colab

◆ 1. Installing Python

✓ Python comes pre-installed in most systems, but here's how to install or check:

For Windows:

- Visit: <https://www.python.org/downloads/>
- Download and run the installer.
- During installation, check ✓ "Add Python to PATH".

Verify Installation (in terminal/command prompt):

```
bash
```

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```
python --version
```

python --version

◆ 2. Installing PyTorch

✓ Recommended via pip (for local machines).

Use this command in terminal (with pip installed):

```
bash
```

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```
pip install torch torchvision torchaudio
```

Verify installation:

```
python
```

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```
import torch
print(torch.__version__)
print("CUDA Available:", torch.cuda.is_available())
```

pip install torch torchvision torchaudio

import torch

print(torch.__version__)

print("CUDA Available:", torch.cuda.is_available()) //this can be avoided

◆ 3. Using Google Colab (No Install Needed)

Google Colab is a cloud-based Jupyter notebook environment with Python & PyTorch pre-installed.

Steps:

1. Go to: <https://colab.research.google.com>
2. Click "New Notebook".

◆ 4. Upload Files to Google Colab

Option 1: Manual Upload

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```
from google.colab import files
uploaded = files.upload()
```

- This will prompt you to upload a file manually.



Option 2: Mount Google Drive

python

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```
from google.colab import drive
drive.mount('/content/drive')
```

- Then access files via `/content/drive/MyDrive/your_folder_name`